Title page

Title

OligoVigil: a curator-verified, source-anchored database of safety and off-target evidence for therapeutic oligonucleotides

Running title (≤50 chars)

OligoVigil: curated ASO/siRNA safety evidence *(48 characters)*

Authors

Jie Ni[1,2,3,*], Xinting Zhang[1], Zhuoying Xie[1], Shan Lu[4], Yun Liu[1,5,*], and Adam Jatowt[2,*]

[1] State Key Laboratory of Digital Medical Engineering, School of Biological Science and Medical Engineering, Southeast University, Nanjing 211102, Jiangsu, China [2] Digital Science Center, University of Innsbruck, Innsbruck 6020, Tirol, Austria [3] Department of Medical Informatics, School of Biomedical Engineering and Informatics, Nanjing Medical University, Nanjing 211166, Jiangsu, China [4] Women and Children Department, The First Affiliated Hospital, Nanjing Medical University, Nanjing 210029, Jiangsu, China [5] Department of Information, The First Affiliated Hospital, Nanjing Medical University, Nanjing 210029, Jiangsu, China

* Corresponding authors.

ORCID identifiers

Author	ORCID
Jie Ni (corresponding)	`0009-0003-9767-5441` — https://orcid.org/0009-0003-9767-5441
Xinting Zhang	`0009-0005-0158-3679` — https://orcid.org/0009-0005-0158-3679
Zhuoying Xie	`0000-0003-3534-1924` — https://orcid.org/0000-0003-3534-1924
Shan Lu	`0009-0004-3088-5070` — https://orcid.org/0009-0004-3088-5070
Yun Liu (corresponding)	`0000-0002-4311-3772` — https://orcid.org/0000-0002-4311-3772
Adam Jatowt (corresponding)	`0000-0001-7235-0665` — https://orcid.org/0000-0001-7235-0665

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Corresponding author contact

Corresponding author	E-mail
Jie Ni (lead corresponding)	njie@seu.edu.cn
Yun Liu	liuyun@njmu.edu.cn
Adam Jatowt	adam.jatowt@uibk.ac.at

Co-author e-mail addresses

Author	E-mail
Xinting Zhang	xtzhang@seu.edu.cn
Zhuoying Xie	zyxie@seu.edu.cn
Shan Lu	lushan_sd@njmu.edu.cn

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Declarations summary

- **Conflict of interest:** The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.
- **Data availability:** Release data (737 curator-verified records: 626 toxicity + 111 off-target; audit trail, benchmark splits, deterministic baselines), schema, API and code are openly available without login at `https://oligovigil.pages.dev/`; the versioned release snapshot is archived at Zenodo DOI `10.5281/zenodo.20633779`. Full statement: `declarations/03_data_availability.md`.
- **Inter-rater agreement:** A second independent curator reviewed a 100-row mixed accept+reject stratified sample (KAPPA-2); binary Cohen κ = **0.34** (fair, per Landis-Koch); the second curator is systematically stricter than the curator of record (a safety-conservative direction). The pilot 100-row KAPPA-1 release-only round (κ not computable; Grade-axis κ = 0.21) is superseded. Full inter-rater treatment in §Methods Stage 3.
- **Code availability:** Curation scripts, QA suites, baseline reproducibility code and the portal source are released under the MIT License (code) and CC BY 4.0 (derived annotations); see `LICENSE` and `LICENSE-DATA` in the repository.
- **Ethics:** No human subjects, animal experiments or identifiable personal data were involved; the resource is derived from published preclinical literature.
- **AI-assisted curation disclosure:** A documented AI-assisted human curation workflow was used (Stage-2 source-grounded LLM proposals with verbatim grounding gate, followed by single-human adjudication); the firewall between machine proposals and human decisions is enforced in the data model and audited per record. See `declarations/06_genai_disclosure.md`.

- **Author contributions (CRediT):** see `declarations/01_credit_author_statement.md`. Summary: **Jie Ni** — Conceptualisation; Methodology; Software; Investigation; Formal analysis; Data curation; Visualisation; Writing — original draft; Project administration. **Xinting Zhang** — Investigation; Data curation; Validation. **Zhuoying Xie** — Investigation; Visualisation. **Shan Lu** — Resources; Writing — review & editing. **Yun Liu** — Conceptualisation; Supervision; Writing — review & editing; Resources. **Adam Jatowt** — Supervision; Writing — review & editing; Resources.

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- Main text (Introduction through Future directions, excluding figure captions, declarations and references): **2,987 words**
- Number of figures: **6** (Figure 1: curation pipeline; Figure 2: evidence-object architecture; Figure 3: evidence landscape; Figure 4: web portal walkthrough; Figure 5: peer comparison; Figure 6: validation dashboard)
- Number of tables: **0**
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